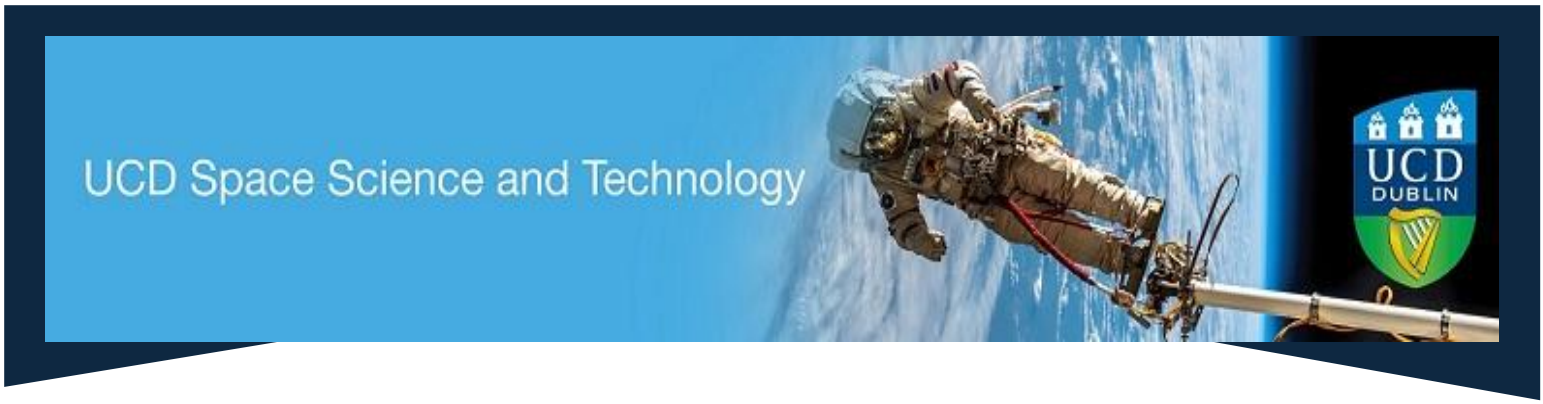


TiNi:

A Gamma-Ray Mission for Measurement of Radioactive Nickel and Titanium in Type Ia Supernovae.

A double-spacecraft Laue Lens observational mission concept for Type Ia Supernovae nucleosynthesis processes.

Orlaith Ní Dhúill

PHYC41100
Space Mission Design Skills

24/04/2026

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1. Declaration of Contributions

Team Supervisor

- Charlotte Bridgett, University of Southampton.

Team Members;

- Chelsea Gamble, (*CG)* - University of Southampton
- David Zazo Callejas, (*DZC)* - University of La Laguna
- Hugo Morin, *(HM)* - University of La Laguna
- Liam Moran, *(LM)* - University College Dublin
- Orlaith Ní Dhúill, *(OND)* - University College Dublin
- Owen Wilson, *(OW)* - University of Southampton
- Riya Kudhail, *(RK)* - University of Southampton
- Sergio Riquelme Salinas, *(SRS)* - University of La Laguna
- Sofia Abad Gómez *(SAG)* - University of La Laguna



Figure 1. TiNi Team Photo while completing the presentation preparation.

1.1 What I did…

Over the week I ended up contributing across many parts of the mission, but the main areas I spent the most time on were the science case, detector decisions, shielding, and later turning everything into a well presented presentation.

At the start of the week, I was mainly focused on understanding the science idea and working out what we were actually trying to learn from the mission through a lot of research with past science papers relating our mission topic of nucleosynthesis. A lot of this was narrowing the science down to Ni-56 in Type Ia supernovae, what the gamma-ray lines could tell us, and why this was still an open problem even after INTEGRAL's many impressive years of data collection.

As the design became more obvious, I got much more involved in the detector side, especially the trade-offs between germanium, CdTe, and scintillator options. A lot of my input here was mainly focused on what was actually realistic for our mission once cooling mass, detector thickness, and shielding were taken into account. I also spent time working through whether the cryocooler assumptions and shield thicknesses made sense for the mission and our set limitations. By the end of the week, once the we worked on and presented our presentation and after that was done, I was mostly focused on pulling all of the noting all of our design choices together logically for the report.

1.2 What the team did…

Even though we had rough subsystem splits at different points, in reality everyone worked on everything in one way or another. The week definitely changed day-by-day. Early on, most of us were still figuring out the science and throwing around different mission ideas, so there was a lot of overlap. As the days went on, individual people started taking the lead on things like lens design, detector choice and sizing, sensitivity, spacecraft mass, and orbit calculations, but all of those decisions were still being constantly discussed as a full group. A lot of the individual mission choices affected everything else, so it never really stayed separated for long. For example, once the lens' focal length choices changed, that affected the detector size, spacecraft separation distances, pointing decisions, and overall mass. The same happened with shielding thickness and cryocooler mass. Because of that, the final TiNi design really came from constant team discussion and shared decision making, rather than each person only doing one section and hoping for the best...

How the week went…

The week felt like it had a 'controlled chaos' flow to it… The Team Supervisors knew it would work out in the end, but we absolutely did not! Between the weather warnings, floods, secret forbidden meetings amongst group members and airport delays its safe to assume we had no idea what each day would bring but I've outlined a general outline of what happened in terms of the mission design process below.

Start of the week

At the beginning it was much more about understanding the science and background of our allocated mission topic, this was done through a lot of research and looking at previous missions, deciding what the mission should focus and mainly getting to know the team members. This was the most open part of the project.

Middle of the week

The middle was where the mission actually started becoming real. This was when we had to stop exploring every idea and start deciding on core things like what we actually wanted to do for our mission and as such; what detector we were using, what energy range we needed, what the field of view should be, if we were to use one… how the Laue lens would work, whether the mass budget still made sense and other major mission questions like these. This was probably the most important and stressful part because it forced everything into one consistent design, which ultimately had to be changed a few times.

End of the week

The last part of the week was mostly about making sure everything actually made sense and could work. This was very stressful, as changing even small things could have a huge butterfly effect to other aspects of the overall mission so a lot of trade-offs had to be considered. That meant checking the Ariane size limits, mass budgets (…so many rechecks)**,** power considerations**,** orbit decisions**,** formation flying analysis**,** cooling and most importantly, whether our science goals still matched the final design. By this point it was less about coming up with new ideas and more about making sure the whole mission made sense as one system, and ensuring we all understood all aspects of the mission and not just what we focused on. Also, a core part of the very end of the week was ensuring we had a good and well thought-out presentation ready to be presented for the morning of the final day.

1.3 Who worked on what… See full name of each contributor in 'Team Members' above

Even though everyone ended up contributing across nearly all parts of the mission, certain people naturally took the lead on different areas as the week went on.

- SAG, CG and OW mainly took the lead on the science case, especially the core nucleosynthesis questions and made sure the final science goals still matched the mission design.
- DZC and LM focused most on the (extremely tricky) Laue lens design, including the crystal layout, focal length, and the formation-flying optical geometry.
- OW and OND (myself) mainly worked on the detector side, including the detector choice, dimensions, cooling assumptions and neccesities, and how the detector linked to the lens and shielding.
- HM and SRS spent most of their time on the coded sensitivity and background work, including the background curves, source fluxes, and making sure the performance claims were actually realistic.
- RK, CG and OND (myself again) mainly worked together on the spacecraft and mission profile, especially the orbit choice, delta_v calculations, mass budget, and power budget considerations.

That said, none of these sections stayed as they were for too long. Most of the important decisions had direct effects on the rest of the mission, so values were constantly being discussed, checked and updated as a full team. By the end of the week, it felt much more like one shared mission design rather than separate pieces of work.

The full presentation is available at: https://tinmission.my.canva.site/full-presentation. This presentation contains some additional figures, intermediate design steps, and supporting material that informed the final mission concept described in this report.

1.4 Whether any writing tools were used…

We used Miro for brainstorming and tracking the mission design as it developed through the week. We also used Google, Google Docs, ChatGPT and Microsoft Word for research, note sharing, and writing, and Canva for the presentation layout and figures. I made sure the final report was written in my own words and matched my understanding of the mission design.

2. Abstract

TiNi is a double-spacecraft gamma-ray mission concept designed to investigate radioactive nucleosynthesis products from Type Ia supernovae and young supernova remnants. The main science goal of our mission is to directly detect gamma-ray lines from the decay of ^{56}Ni and ^{56}Co, which can be used to better constrain the amount, distribution, and evolution of radioactive material within the ejecta.

A secondary science objective is the detection of ^{44}Ti emission from young supernovae remnants, helping to extend our current limited scientific knowledge beyond transient Type Ia events. The mission concept uses an 86m formation-flying Laue lens to focus gamma rays onto a compact cryogenically cooled germanium detector, chosen to maximise sensitivity to narrow gamma-ray lines in the MeV range. An active and passive shielding system including a BGO shield and a sun-shield is included to reduce instrumental background noise and improve line detectability.

The spacecraft pair is designed to fly and operate at a Sun - Earth L2 posiotion, where long uninterrupted observations, reliably stable thermal conditions, and safe formation flying are possible over an 8-year mission lifetime. The mission builds on the heritage of previous gamma-ray observatories like INTEGRAL, while aiming to improve sensitivity to key radioactive supernova lines. Overall, TiNi is designed to help answer questions that are still very open around Type Ia progenitor models, ejecta structure, and radioactive line escape.

3. Introduction

Nucleosynthesis is one of the most important processes in astrophysics because it explains how the elements are formed and distributed throughout our Universe. Many of the heavier elements found in stars, planets, and even within the Solar System are produced during stellar evolution and explosive events. Understanding where these elements come from, how much is produced, and how they are transported into the interstellar medium remains a key part of understanding both stellar life cycles and galactic chemical evolution. Within this, Type Ia supernovae are especially important because they are one of the main sources of iron-group elements, particularly ^{56}Ni, which is synthesised during the explosion. The radioactive decay chain '^{56}Ni to ^{56}Co to ^{56}Fe' is really important to the physics of the event, as the decay energy powers the optical light curve observed after explosion. Measuring this radioactive material directly gives a much clearer view of what is happening inside the ejecta than in optical observations alone. This is where gamma-ray observations are pretty uniquely powerful, interesting and useful for our current understanding. The decay of ^{56}Co produces gamma-ray lines at well-defined energies, which act as direct signatures and clues of the radioactive material created in the explosion. Unlike optical emission, which is reprocessed through the expanding ejecta, these gamma rays provide a much more direct probe of the amount, distribution, and escape velocity of the radioactive products. Although INTEGRAL has shown that direct detection of these lines is possible, only a very small number of nearby supernovae have actually been observed this way. This means that major questions are still very much so open, including how much ^{56}Ni is produced, how it is distributed through the ejecta, and how the gamma-ray escape evolves over time. The limited sensitivity of previous missions means that these direct detections are currently restricted to only the nearest and brightest events. This motivates the need for a more sensitive gamma-ray mission capable of improving direct line measurements and placing stronger constraints on Type Ia progenitor models, explosion asymmetries, and ejecta structure**.**

4. Science Case

Type Ia supernovae are one of the main nucleosynthesis sites for the production of iron-group elements, particularly ^{56}Ni, which is synthesised during the thermonuclear explosion of a white dwarf. The radioactive decay of this nickel powers much of the observed supernova light curve and provides one of the clearest ways to probe the explosion physics directly. The decay chain is: ^{56}Ni to ^{56}Co to ^{56}Fe. The early 812 keV ^{56}Ni line and the later 847keV and 1238keV ^{56}Co lines give clues of the radioactive material within the ejecta. Unlike optical observations, which depend on radiation reprocessing through the expanding supernova envelope, gamma-ray line measurements allow a much more direct view of the radioactive yield, line broadening, and transparency evolution of the explosion.

4.2 Main science question

The main science question driving the TiNi mission is:

Through observing the evolution of the ^{56}Ni energy line and decay curve produced by Type Ia supernovae, how well can we identify the progenitor system and define the mechanics of the explosion?

More specifically, the mission aims to test whether gamma-ray line evolution can distinguish between different explosion scenarios, including:

- Single Degenerate Progenitors
- Double Degenerate Progenitors
- ‘Deflagration’ vs ‘Detonation’ dominated models
- Delayed detonation scenarios

The shape of the line, its width, the rise to peak, and the decay timescale all provide information on the amount of ^{56}Ni produced and how asymmetrically it is distributed through the ejecta. Our team mission hopes to use this as a way to distinguish between narrow emission line models and broader asymmetric ejecta models, which links instrument design to the big progenitor question.

4.3 What earlier missions have taught us…

Previous missions, especially the INTEGRAL mission, has shown us that direct detection of radioactive supernova gamma-ray lines is possible, especially for rare nearby events like SN2014J. This proved that gamma-ray spectroscopy can directly look at supernova nucleosynthesis and line escape. However, these detections were limited to the nearest events, which is why our mission extends the observation radius from the 5Mpc which was used by INTEGRAL to 9Mpc. This improves the expected detection rate to approximately 0.26 Type Ia supernovae per year, giving a hopeful, yet realistic expectation of around 2 supernovae during the 8-year mission lifetime.

The main limitations from previous missions are still:

- Insufficient sensitivity in the MeV range
- Pretty strong instrumental background noise.
- Too little directly observed Type Ia events constrain models statistically.

4.4 What is still unresolved

Despite previous detections, research and missions, several huge questions are still unanswered:

- How much ^{56}Ni is actually produced in different Type Ia explosions?

- Can line broadening help us to properly distinguish between single and double degenerate progenitors?
- How mixed or asymmetric is the ejecta? What does this tell us if so?
- At what point does the supernova become transparent to gamma rays?
- Can the rise and decay timescale of the line constrain the explosion model?

These open questions form the main scientific justification for TiNi. A secondary science case is the observation of ^{44}Ti emission at 1157 keV from young Galactic supernova remnants such as G1.9+0.3 and Kepler's supernova remnant, which would allow the mission to continue observing and collecting nucleosynthesis based science data between nearby Type Ia detections.

5. Quantitative requirements

The science goals described in *Section 2* are core to the key instrument and mission requirements for TiNi. The primary requirement is to detect the evolution of the 812keV ^{56}Ni line and the later 847keV and 1238keV ^{56}Co decay lines from nearby Type Ia supernovae, while maintaining optimal sensitivity to observe secondary 1157keV ^{44}Ti emission from young Galactic remnants.
The main quantitative requirements are summarised in 'Table 1' below.

Table 1. Requirements of TiNi mission with reasoning.

Requirement	**Value**	**Reasoning**
Primary science lines	812, 847, 1238 keV	Direct ^{56}Ni, ^{56}Co decay signatures
Secondary science line	1157 keV	^{44}Ti emission from young SNRs
Energy band	800 - 1300 keV	Covers all primary and secondary lines
Energy resolution	0.27 - 0.77%	Required to resolve line width and broadening
Observation type	Pointed follow-up	SNe Ia identified by optical discovery
Field of view	1 arcmin	Driven by Laue lens design
Mission lifetime	8 years	Expected to capture approx. 2 nearby SNe Ia
Orbit	Sun - Earth L2	Long observing time, low background, stable thermal conditions
Primary target distance	9 Mpc	Extended from INTEGRAL 5Mpc
Expected SNe Ia rate	0.26 yr	2 events over mission lifetime
Primary observation time	100 days per SNe Ia	Track line rise, peak, and decay
Secondary observation time	180 days per year	Long integrations on ^{44}Ti remnants
Timing cadence	Repeated monitoring over 100 days	Track transparency and ejecta evolution

Flux assumptions

The expected line flux assumptions used in the mission design are based on scaling from previous missions and nearby source distances.

- For the primary science case, the expected ^{56}Ni flux at 9Mpc is approximately: $1.2\text{x}10^{-5}$ ph/cm^{-2}/s^{-1}
- For the secondary ^{44}Ti science case at 8kpc, the expected flux is: $1.2\text{x}10^{-6}$ ph/cm^{-2}/s^{-1}

These values are used as the main sensitivity parameters for the detector, shielding, and lens design.

Requirement assumptions

Some subsystem-level values, particularly source occurrence statistics and secondary target observation cadence, were estimated using a combination of field-trip calculations, mission heritage from INTEGRAL/MAX, and first-order scaling assumptions. Due to time limitations during the field-trip design exercise, some values were fixed using past mission requirements and reasonable engineering estimates rather than fully accurate and perfectly calculated requirement estimates. These assumptions are not '*set-in-stone'* and should be considered entirely assumption based only!

6. Instrument overview

The TiNi payload is designed as a pointed gamma-ray line spectrometer optimised for direct observation of radioactive supernova decay lines in the 800 to 1300 keV range. The telescope concept combines four main elements:

1) A Laue lens as the primary focusing system
2) A minimum-mass HPGe focal-plane detector
3) An active BGO veto shield
4) A double spacecraft positioned in formation-flying long focal-length geometry

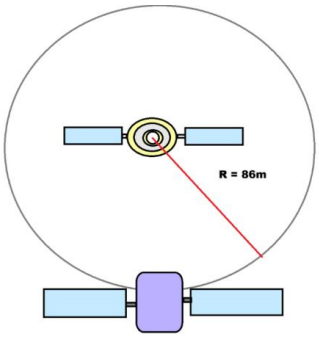


Figure 2. Formation Flying of Double Spacecraft with 86m separation.

The Laue lens concentrates the incoming gamma-ray flux from the target line energies onto a compact detector area, allowing the instrument to maximise discrete-line sensitivity while minimising detector background volume. This is especially important for the low expected line fluxes from nearby Type Ia supernovae. As such, the payload operates as a highly sensitive pointed follow-up spectrometer, with source positions supplied by optical supernova alerts.

Laue Lens Parameters

The Laue lens focuses high-energy gamma rays using Bragg diffraction, where photon energy is related to diffraction angle by:

$$2d_{hkl}sin\left(\theta\right) = n\lambda = n\frac{hc}{E}$$

Different energies are diffracted at different angles, so the lens is arranged in two concentric rings targeting the ^{56}Ni or 812keV and ^{44}Ti at 1157keV emission lines. For optimal alignment the system would need a focal length of 86m of separation, this meant the mission would require formation flying between

the lens segment and detector segment of the full spacecraft payload. The lens consists of concentric crystal rings with radii between 38.44 cm and 59.62 cm, this allowed for simultaneous focusing of multiple energy bands. Gold mosaic crystals are used for their high diffraction efficiency. The mosaic spread ($\Delta\theta$) determines the energy bandwidth and focal spread, with the focal radius (FR) calculated by: FR = $\Delta\theta$ x (f). For ^{56}Ni, $\Delta\theta = 10''$ and FR = 0.46 cm, while for ^{44}Ti, $\Delta\theta = 5''$ giving FR = 0.23 cm. A crystal thickness of 1mm was picked for TiNi, resulting in a reflection efficiency of approximately 42%. Overall, the lens design balances energy coverage and focusing precision to optimise gamma-ray detection.

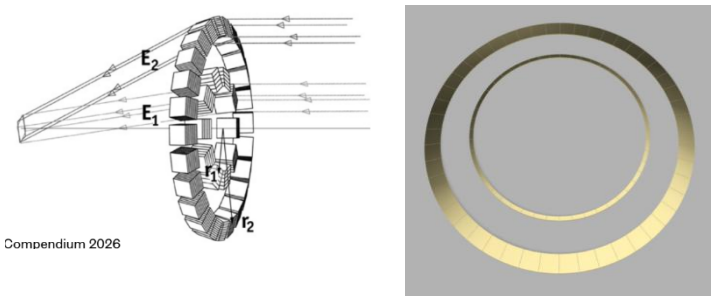


Figure 3. Laue Lens Concept Design (Material: Gold, as its most efficient for Bragg Diffraction)

6.2 Why this concept was chosen…

This instrument concept was selected because the science case is fundamentally line-sensitivity limited rather than imaging limited. The use of a Laue lens provides the largest performance gain by focusing a large effective collection area onto a very small focal-plane detector. This improves narrow-line sensitivity by reducing the detector volume exposed to diffuse and activation background, which unfortunately is one of the main limitations in MeV gamma-ray astronomy. A minimal HPGe detector was chosen to preserve the excellent spectral resolution which we would need for radioactive line measurements, while keeping the cryocooler and shielding mass as low as possible. This was an important trade throughout the design process, as a larger germanium detector size would improve stopping power but quickly increase cryo cooling mass and the overall spacecraft complexity. An active veto shield is needed because the instrument would be badly affected by background, especially from cosmic diffuse gamma rays, secondary activation, and shield leakage. Without active shielding, the line sensitivity gain from the Laue lens would be far lower.

6.3 Detector choice

The final detector choice is a compact coaxial high-purity germanium (HPGe) focal-plane detector. This was selected because the primary science lines at 812keV, 847keV, 1157keV, and 1238keV all sit close to the 1 MeV regime, where germanium detects at optimal spectral resolution. The detector was intentionally kept as small as possible while still maintaining sufficient stopping power for these line energies, based on the handbook detector restrictions but also for feasibility. A CdTe detector was also very briefly considered during the design phase because it offers room-temperature operation and reduced cooling complexity. However, for the required energy range it becomes significantly less practical due to limited crystal thickness, the need for multiple stacked layers, and charge trapping effects that degrade line performance. A scintillator-only detector was also really briefly considered as a lower-mass option, but this was rejected because the reduced spectral resolution would weaken the mission's ability to

measure line width, Doppler broadening, and transparency evolution, all of which are pretty important to the science case. For this reason, the compact HPGe option was the best balance between spectral performance and spacecraft mass constraints.

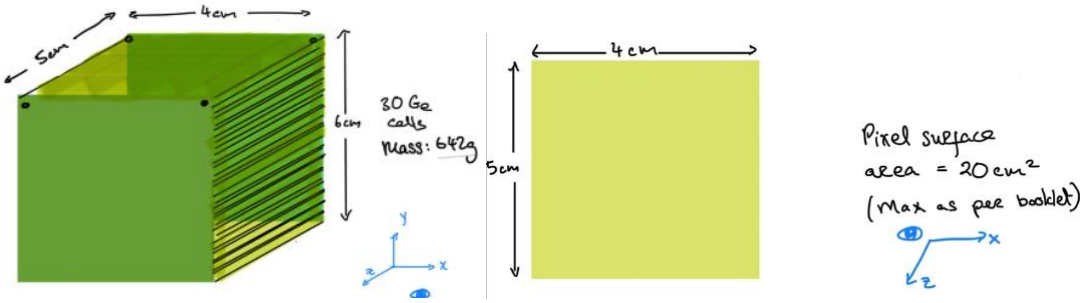


Figure 4. HPGe Detector Design Diagram with dimensions.

6.4 Shielding

Active shielding was selected as an essential part of the payload. The concept uses an anti-coincidence BGO shield surrounding the focal-plane detector, with a lightweight passive outer cap inspired by the INTEGRAL SPI and Fermi anti-coincidence approach. This shielding geometry is intended to reject charged particle events, diffuse off-axis gamma rays, local activation products, secondary background generated in surrounding spacecraft structure. Additionally, we opted for an external sunshield and plastic top for a reliable passive shielding source.

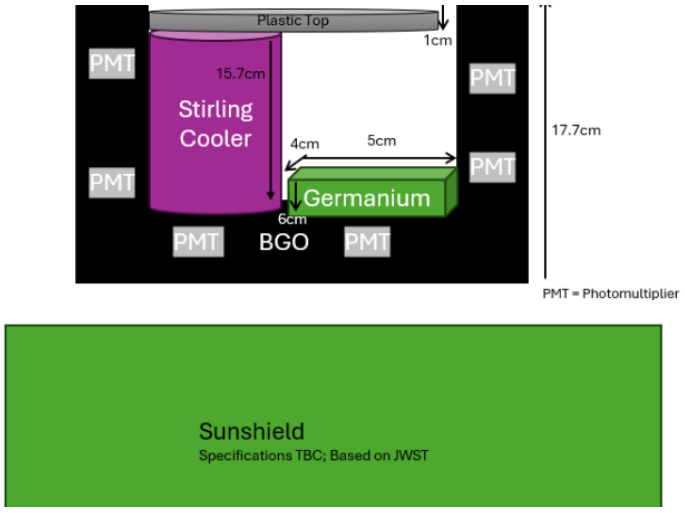


Figure 5. Full early payload diagram, with BGO active shield and 6 photomultipliers. See also the early concept sunshield and plastic cover.

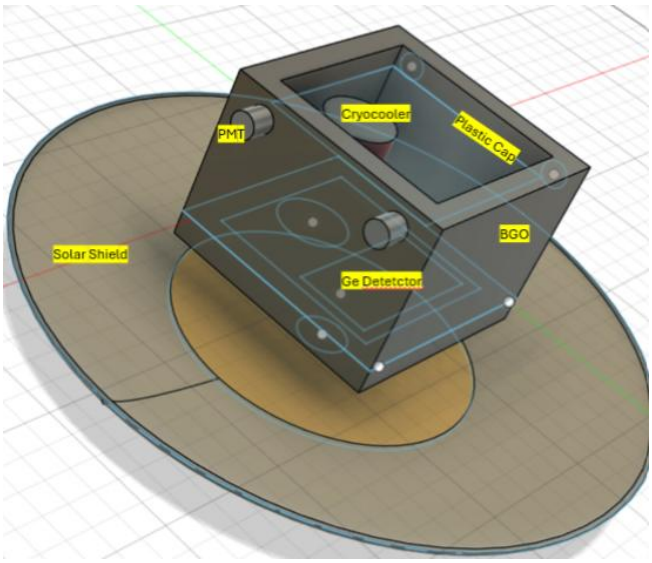


Figure 6. Later, more mature CAD diagram of detector side payload.

7. Instrument analysis

7.1 Sensitivity

Sensitivity is the main performance parameter need for TiNi. The mission is designed to detect very faint radioactive gamma-ray lines from nearby Type Ia supernovae, so the limiting factor is not whether the detector can measure photons at these energies, but whether it can do so above the combined instrumental and diffuse background. This is a standard difficulty in gamma-ray astronomy, where the background often dominates over the astrophysical signal.

For TiNi, this is especially important because the expected ^{56}Ni and ^{56}Co line fluxes are low even for nearby sources, meaning the mission is very background-limited rather than source-limited. Realistically, this means that any increase in detector background, shield leakage, or activation can have a big overall impact on the final line detectability.

For this reason, discrete line sensitivity is more important to this mission than continuum sensitivity. The science goals depend on detecting and characterising specific radioactive decay lines, particularly at 812keV, 847keV, 1157keV, and 1238keV, rather than measuring broad-band spectra.

The payload was therefore designed to maximise narrow-line sensitivity through the combination of:

- a Laue lens to focus photons onto a small detector area,
- a HPGe detector to preserve line resolution,
- Both passive and active shielding to reduce background events.

The mission sensitivity estimates used during the field trip mission design were based on first-order assumptions for:

- Detector efficiency as a function of energy,
- Background contributions from diffuse cosmic background, activation/spallation, and shield leakage.
- a 3σ detection threshold.

These values were then compared to the instrument sensitivity targets developed during the mission trade studies. The resulting performance was intended to improve on older missions such as INTEGRAL, while remaining broadly consistent with the higher-sensitivity focused gamma-ray approach proposed in the MAX mission concept. Due to time limitations during the field trip, the sensitivity treatment remained at first-order design level rather than a fully optimised end-to-end simulation. The purpose of the analysis was therefore to show that the concept was plausible and internally consistent, rather than fully flight-qualified.

7.2 Spectral performance

The spectral performance of the payload is central to the science case. Since the main objective is to detect and interpret narrow radioactive decay lines, the detector must preserve sufficient energy resolution to distinguish line peaks, broadening, and line evolution over time. A compact HPGe detector was chosen because germanium provides the best spectral resolution available for the target energy range around 1MeV. This is particularly important for the 847keV and 1238keV ^{56}Co lines, where the line profile contains physical information about the ejecta, including Doppler broadening and the distribution of radioactive material.

The detector thickness was chosen as a compromise between:

- Improved stopping power at high energy
- Detector mass
- Cryocooler mass
- Total shielding volume

A thicker detector would improve high-energy interaction efficiency but would also increase the cooling stored mass and total payload mass. The final concept therefore used a minimal HPGe focal-plane detector, accepting some efficiency trade-off in order to maintain a more realistic spacecraft design. Cooling also directly affects spectral performance. Since HPGe can't really operate properly at room temperature, the cryogenic system is not optional: it is absolutely needed and essentail for maintaining the energy resolution needed for narrow-line spectroscopy. This was one of the main reasons the detector was kept small compact, also the decision on this was heavily influenced by the notebook restrictions on detector size, area and cooling requirements.

7.3 Temporal performance

TiNi is not a fast-timing mission. Its temporal requirements are based on the physical evolution of supernova line emission rather than by sub-second variability. For the primary science case, the important

timing requirement is the ability to carry out repeated observations over approximately 100 days after the detection of a nearby Type Ia supernova.

This allows the mission to follow the:

- Appearance of the line
- Rise to peak
- Decay after maximum
- Changing transparency of the ejecta

7.4 Angular performance & field of view

The angular performance of TiNi is driven by the use of a Laue lens, which naturally gives a small field of view. This is acceptable for the mission because TiNi is not meant to discover transients independently. Instead, it is designed to perform follow-up observations of sources whose positions are already known from optical surveys and supernova alerts. The narrow field of view is therefore not a major disadvantage. In fact, it is part of the trade that allows the mission to gain sensitivity by concentrating the desired line photons onto a small detector area. This means the payload should be understood as a high-sensitivity pointed spectrometer, not a wide-field survey instrument. The small FoV is therefore a direct consequence of the mission science priorities rather than a design weakness.

8. Mission design / spacecraft design

8.1 Orbit

TiNi is designed to operate at Sun-Earth L2. This was chosen over LEO and HEO because it best supports the science and instrument requirements of the mission. The main advantage of L2 is that it allows long, uninterrupted observations, which is essential for a mission focused on faint radioactive gamma-ray lines from nearby Type Ia supernovae. Since the science depends on tracking the evolution of line emission over timescales of days to months, frequent Earth occultation and shorter observing windows in LEO would significantly reduce observing efficiency. By contrast, L2 allows the mission to carry out much longer continuous integrations on the same source. A second major advantage is the radiation environment. At L2, the spacecraft is well outside the Earth's trapped proton and electron belts, reducing activation effects and lowering the instrumental background compared with lower Earth orbits. This is especially important for TiNi because the mission is background-limited, and even small increases in activation or shielding leakage would directly reduce line sensitivity. L2 also suits the overall mission architecture. The payload uses a long focal-length Laue lens, requiring two spacecraft flying in formation approximately 86 m apart, and this is easier to maintain in a more stable environment than in low Earth orbit. In addition, the thermal environment at L2 is more stable, which benefits both the detector cooling system and spacecraft operations over the planned 8-year mission lifetime. For these reasons, L2 was selected as the most suitable orbit for TiNi, despite the increased transfer requirements compared with LEO. The total mission Δv budget used in the design was approximately 12.19 km s^{-1}, which remains consistent with the mission launch and transfer assumptions used during the field-trip design exercise.

8.2 Mission lifetime

A 5-year baseline lifetime is considered the minimum scientifically useful mission duration for TiNi, with the mission designed to be around an 8-year nominal lifetime and a 10-year stretch goal. The main reason for this is the rarity of nearby Type Ia supernovae. Based on the expected event rate within 9 Mpc, an 8-year mission gives a realistic chance of observing around 2 nearby events, which is enough to begin comparing progenitor and explosion models. A shorter mission would make it much less likely that enough suitable supernovae would actually occur during the operational phase. From the engineering side, this lifetime is realistic because similar HPGe detectors and cryocooler systems have worked well in past long-duration gamma-ray missions, making a 5-10 year operational window totally realistic and doable. The 10-year stretch goal mainly improves the chance of additional nearby supernova detections and increases the secondary science return from ^{44}Ti remnant observations.

8.3 Spacecraft configuration

The TiNi payload is based on a two-spacecraft formation-flying configuration, driven directly by the 86m focal length required by the Laue lens. One spacecraft carries the lens assembly, while the second carries the focal-plane detector, shielding, and cryogenic systems. This separation is necessary because the focal length is too large to be realistically fit inside a single spacecraft structure. A formation-flying setup gives a way more practical way to preserve the optical performance of the lens while keeping each spacecraft mechanically simple. The detector spacecraft houses the compact HPGe focal plane, active BGO veto shield, and the cryocooler system, while the lens spacecraft is mainly responsible for maintaining the crystal alignment and pointing stability required for narrow-line observations. A deployable boom concept was briefly considered as an alternative, but this was not developed further because the two-spacecraft idea worked best for the long focal-length geometry and allowed for better pointing and mass conservation.

8.4 Mass budget

We mainly prioritised the mass budget within the provided notebook and based many mass decisions on staying well within that limit with margin accounted for. The in Table 2 is based on a combination of team design values, past mission assumption scaling, and reasoned estimates

Table 2. Mass Budget approximations

Subsystem	Mass (kg)
Laue lens structure	174
Detector assembly	180
Sterling Cryocooler	25
Shielding (BGO + passive cap)	305
Electronics	100
Power subsystem	300
Communications	25
ADCS / formation control	100
Structure / harness	300
Subtotal	**1509**
15% system margin	**226**
Payload fuel / transfer allowance	**430**
Total	**2165**

The 15% system margin was included to reflect the early mission-concept level of the design and the fact that several subsystem values, particularly shielding, formation-flying control, and structural support, were still based on assumptions. Where exact mass values were not available from the field-trip handbook, educated estimates were used based on the total spacecraft mass and online mass sources from previous gamma-ray missions. The mass budget should therefore be understood as a mission concept estimate rather than a final engineering closure.

9. Trade-offs and limitations

Although the TiNi mission hopes to achieve new and informative science results for better understanding radioactive supernova line observations, several important trade-offs and limitations became clear during the design process.

In my opinion, the most impactful detector trade-off is the use of HPGe detector choice. Germanium provides the best spectral resolution for the 847keV and 1238keV ^{56}Co lines, which is obviously essential for measuring line broadening and ejecta evolution. However, this comes at the cost of a heavy cryocooler, increased shielding volume, and a much higher overall spacecraft mass. A lower-mass detector material such as CdTe would be great as it would mitigate the cooling problems but would also compromise the spectral performance which is most important to the mission science.

The use of a Laue lens is one of the best strengths of our mission concept because it significantly improves line sensitivity by focusing the flux onto a small detector area. That said, this also introduces major problems and creates payload complexity. The narrow field of view means the mission depends entirely on external optical discovery of nearby Type Ia supernovae, and the long focal length drives the need for a two-spacecraft that are appropriately formation-flying. This, of course, really increases the mission risk compared with a more conventional single-spacecraft payload.

Another big limitation of this mission is the energy dependence of the lens performance. Thought the lens is well suited to the primary 847keV line, the higher 1238keV line may be more difficult to optimally ‘see’ simultaneously in the same narrow-band crystal geometry. This means that there is likely some performance trade-off between the two main science lines, which is unavoidable unless a more complex and better suited multi-ring lens is brought into the mission concept.

One of the main considerations worth mentioning is that the science results are also extremely dependent on the mission lifetime and source occurrence statistics. Because the mission relies on nearby Type Ia supernovae occurring within the 9Mpc search window, the number of useful primary science source targets during the nominal lifetime is very small and I’m not a betting woman so I’m not sure I like the odds for such an expensive mission.

Finally, background modelling remains one of the largest uncertainties in the mission concept. Although the combination of focused optics and active shielding should improve narrow-line detectability, the final sensitivity still depends heavily on activation background, shield leakage, and local spacecraft-induced events. A more detailed background simulation would therefore be one of the most important next steps in a future design phase. Overall, the main trade-offs in TiNi are between spectral performance, spacecraft complexity, and mission opportunity statistics, and these limitations are closely linked to the mission’s highly specialised science goals.

10. Improvements / changes on reflection

The TiNi concept reached a strong mission-level design during the field trip, but several areas would clearly benefit from more detailed refinements and improvments. One of the most important improvements would be further optimisation of the Laue lens energy response. In the current design, the lens is required to support both the 847keV and 1238keV ^{56}Co lines, as well as the secondary 1157keV ^{44}Ti line. A future design could either optimise the lens more strongly around the 847keV primary science line, or try to redesign the lens to improve simultaneous performance across all of the desired energies. The shielding design would also need some more educated redesigns. Although the active BGO veto concept is a great starting point, the exact thickness, geometry, and passive outer layers were quick math assumptions during the design stage, but I think they would definitely need to be reconsidered and maybe redesigned if necessary. A more educated and detailed design would improve the balance between background rejection and mass growth as the BGO is very heavy and we found it to be very tricky to perfect in early designs. Although a HPGe detector is obviously the best option for spectral performance and best results, it does also constrain a few things in the overall mission such as the cryocooler mass and mission lifetime, which really isn't ideal for the mission objectives and lifetime, that said cryocooled germanium detectors have outlived our mission lifetime by many years in past missions such as INTEGRAL, RHESSI, XEUS and ROSETTA with a combined average lifespan of 18.75years, well over the TiNi mission stretch goal.

The mass budget should also be refined to be more precise and better than our current assumption based masses, especially for the formation-flying subsystems, structural supports, and shielding mass which were all calculated in a rushed way late on the final night before the presentation. These values are currently educated guesses but would need to be relooked at. Finally, the most important technical improvement would be better refined sensitivity calculations, like: detector response, background noise, activation modelling, shield leakage and also line detectability as a function of observation time.

11. Teamwork and division of labour

The TiNi mission concept was developed as a team over the course of the full field-trip week. Although rough subsystem leads were formed as the design became more detailed, the work remained highly shared across the team. Broadly, effort was distributed across:

- Science case development
- Lens and focal geometry
- Detector and shielding design
- Sensitivity and background calculations
- Spacecraft, orbit, and mass budgeting

Most design decisions were made through regular full-team discussions, where changes in one subsystem were immediately checked against their effects on the rest of the mission. This was especially important for parameters such as focal length, detector size, shielding mass, and mission lifetime, which all directly affected the wider spacecraft design. My own strongest contribution was early in the science case development as we chose to follow my 'Big Question', I also worked most on the detector background and all of the detector trade-offs, shielding choice and logistics, and ensuring the final presentation clearly reflected the TiNi mission goals and reasoning behind the design choices. This workflow was one of the

strengths of the project, as it allowed the mission to evolve into a consistent end-to-end concept instead of a set of disconnected subsystem designs.

12. Conclusion

TiNi is proposed as a dual-spacecraft gamma-ray mission designed to directly study radioactive nucleosynthesis products from Type Ia supernovae and young supernova remnants. The mission is mainly focused on the detection of the 812keV ^{56}Ni line, the later 847keV and 1238keV ^{56}Co lines, and secondary 1157 keV ^{44}Ti emission data collcetion, with the aim of improving constraints on progenitor models, ejecta structure, and the evolution of radioactive line escape. The chosen instrument concept, combining a formation-flying Laue lens, compact HPGe focal-plane detector, and both active BGO shielding and passive sunshielding, is well suited to this science case because it prioritises narrow-line sensitivity in the MeV range, which is the main driver of the mission. The orbit, lifetime, and formation-flying set-up were all chosen to support long uninterrupted observations of rare nearby Type Ia events. The main limitations of the mission proposal are the dependence on the occurrence of sufficiently nearby supernovae, the complexity of the long focal-length two-spacecraft design, and the remaining uncertainty in detailed background modelling. Despite these challenges, the mission is still really scientifically valuable because it directly targets several unresolved questions in Type Ia explosion physics that can’t be answered as cleanly through optical observations alone. Overall, TiNi builds on the legacy of INTEGRAL by improving narrow-line sensitivity to radioactive supernova nucleosynthesis, while demonstrating a realistic pathway toward a more advanced focused gamma-ray observatory.